Disclosure Day

How can we prove that we're most likely inside an Observer Patch Holography simulation?

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Mathematical certainty belongs to theorem proving. Empirical physics works differently. A theory earns trust when a small set of principles emits a large amount of observed structure without reading the answers from measurement. The stronger version also says where it can fail.

This note makes that case for Observer Patch Holography. OPH starts with finite observer patches, five axioms, and two closure coordinates. Those two coordinates are not free variables. They are fixed points of the local pixel loop and the global screen-capacity loop. Let the axioms and the closures run, and the construction unfolds into a universe: observer records, public spacetime, Lorentz structure with the three-dimensional H^3 spatial chart, Einstein structure, compact gauge reconstruction, the Standard Model quotient, the charged-family shape, the weak hierarchy, gravity normalization, and frozen particle and cosmology tests.

This short paper has one main goal. It shows a small load-bearing slice of the structure and numerical values of our universe from practically nothing: the OPH axioms plus the two solutions of the self-referential closure equations. One equation makes the local pixel readback compatible between the simulating side, the outer readback, and the simulated side, the inner readout. The other does the same for global screen capacity. From that starting point, the slice emits the Standard Model branch integers, the Koide charged-family shape, the QCD-free hierarchy witness, and dimensionless gravity normalization. It looks like our universe in places where accidental fits are cheap, so the anti-fitting rule below says exactly what counts. The note ends by pointing to the detailed papers that show how OPH naturally and elegantly solves hard problems in contemporary physics. Every theorem or proof omitted here is referenced where it is used. Familiar numbers such as public $\alpha(0)$, SI G, measured hadronic inputs, selected PDG frames, cosmology scorecards, hardware lanes, and bridge-defined capacities are comparison or falsification data unless their own calculations are supplied without reading the measured answer. We believe any reasonable reader will conclude that the output universe fits our experience too well to be mere coincidence.

What this proof contributes

The proof compresses the OPH stack into one readable path. Standard tools such as modular theory, fixed-point analysis, entropy gravity, compact gauge reconstruction, and ordinary comparison data are used with their usual meanings. OPH adds the finite observer-patch closure, the two fixed-point coordinates P_{src} and $N_\star$, and the rule that a numerical row counts only when its endpoint is emitted by the declared map. On that basis the note collects the dark-energy capacity closure, the H^3 spatial chart, the Standard Model branch integers, the hierarchy witness, the Koide-family shape, the Newton normalization row, and the main failure tests in one short argument.

Anti-fitting rule

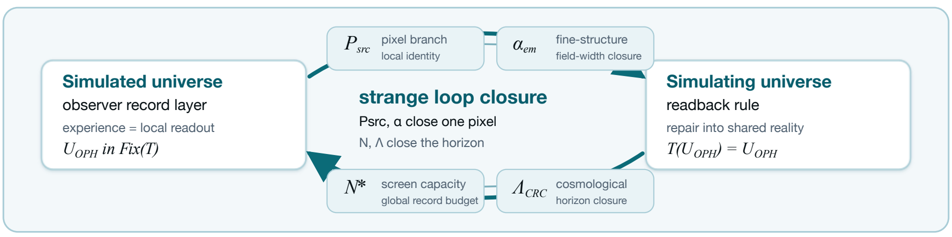
A numerical row is counted in our argument only if the paper or code gives four things: the calculation that emits it, no hidden path from the measured answer into that calculation, an interval or uniqueness check, and a clear way for the row to fail. A measurement may identify the basin to inspect. It may not define the endpoint. If the target value, or an algebraically equivalent calibrated proxy, appears in the calculation path, the row is comparison data, a display check, empirical closure, or a test surface. It is not part of the compact anti-circularity proof.

Most people picture a simulation as a machine that renders a world frame by frame. OPH uses a different mechanism. It does not simulate the contents of spacetime tick by tick on a global timeline. In the fundamental description, the global timeline is absent. The pattern is a fixed-point computation:

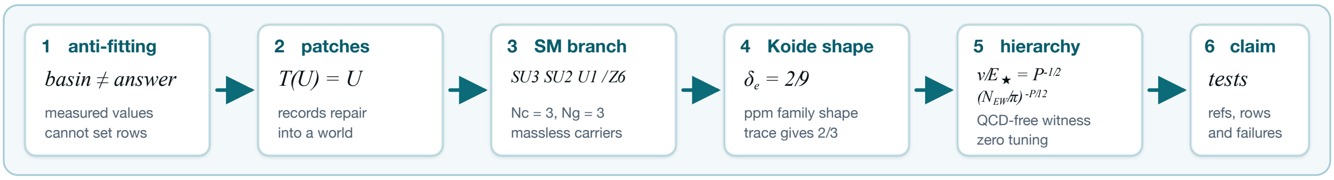
$$\text{ordinary simulation: } U_t \to U_{t+1} \to U_{t+2} \to \cdots, \qquad \text{OPH: } \mathcal{T}(W) = W.$$

The symbol W is the globally self-consistent observer-readable structure. $\mathcal{T}$ is the OPH readback-and-repair operator over observer patches, records, overlaps, mismatch scores, repair moves, and checkpoint continuation [O1,O2,O3]. The universe is the normal form of that computation. From inside, observers read it as history and experience that reading as time. Hofstadter's name for this kind of thing is a strange loop: a "paradoxical level-crossing feedback loop" [S16]. In OPH, strange loops sit at the core of how the universe works. We fully realize that this sounds insane, but this is exactly where rigorous mathematics leads us. Measurements may point to the right branch. Their role is branch localization. If OPH is true, our exact universe, without exception, must emerge from the OPH closure coordinates and axioms [O1,O2,O3,O4,O5,O6,S12,S13]. The counted rows below use that rule for the Koide shape and hierarchy witnesses. The remaining rows are comparison data or falsification surfaces with declared claim classes.

Bracketed source keys matter. When this note names a theorem without printing the full proof, the numbered OPH paper carries the proof details. The direct reading anchors at the end point to the online source lines. For fast lookup: OPH axioms are [A1], consensus repair is [A2], pixel closure is [A4], capacity closure is [A5], the QCD-free hierarchy witness is [A10], and the Standard Model gauge quotient is [A21]. The derivation chains printed here and in the referenced papers should be sound and complete; if you find an error, contact info@floatingpragma.io.



The labels name what the values do: P_{src} fixes the local pixel branch used in the compact witnesses, α is the field-width readback, $N_\star$ fixes total screen capacity, and Λ is the horizon closure seen in cosmology.



OPH's five axioms are listed in *Observers Are All You Need* [O1]. Think of them as the toolchain. The argument below uses the finite observer-patch normal form, the exact quantum event surface, the two closure coordinates, the recovered Standard Model branch, and one compact three-scalar cross-lock:

$$(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4) \quad \Longrightarrow \quad \left(\frac{v}{E_\star}, \frac{m_\mu}{m_e}, \frac{m_\tau}{m_e}\right).$$

The table says which rows are counted in this compact argument and which rows stay comparison data or falsification targets [O1,O3,O4,O5,O7,O8,O9].

1. The fixed-point computation

OPH does not use "simulation" as a synonym for "fixed point." The theorem-level criterion is the OPH simulation certificate [O2,A23]. It requires five independent properties: endogenous update, observer-readable records, overlap repair with a schedule-independent normal form, selected-branch elimination, and implementation/clock closure. The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form after readback [O1,O2,O3]. Internal observers experience that normal form as the universe. Operationally, the certified system consists of finite patches, shared overlap ports, recovery-derived repair maps, checkpoints, and a stable world read from inside [O1,O2,O3].

The symbol $\mathfrak{U}_{\mathrm{OPH}}$ denotes the selected OPH universe. The map T is the observer-overlap repair map [O1,O2]. The notation $\mathrm{Fix}(T)$ means the states returned unchanged by that map:

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

The fixed-point equation is a consequence, not the definition. The selected-packet theorem [O2,A23] proves the five clauses on the declared finite branch and then derives this identity. Its identity-map/constant-functional counterexample shows that an ordinary equilibrium or variational solution does not pass automatically, and [O2,A23] lists structural ways the certificate can fail. Direct lookup in the consensus paper: Definition `def:oph-simulation` states the certificate, Theorem `thm:oph-simulation` proves the selected-packet result, and Remark `rem:oph-simulation-falsifiers` gives the falsifiers. This compact note cites that package instead of repeating its proof.

Read the equation literally. The simulated record world enters the simulating readback rule, contradictory branches die, and the returned object is the same record world. That is the self-contained consistency computation [O1,O2]. A universe with unexplained external constants fails this test [O1,O3,O4]. The finite OPH patch-carrier pipeline is the concrete fixed-cutoff carrier for this computation: a finite patch system with ports, central records, repair moves, checkpoints, and a terminal normal form. It carries the closure solve; it is not a frame renderer.

The observer lock changes the meaning of experience. An observer in OPH is a finite record boundary with a comparison rule for nearby patches [O1,O2]. Subjective experience is that boundary reading itself from the inside:

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\mathrm{OPH}}), \qquad \mathfrak{U}_{\mathrm{OPH}} = T(\{r_i\}_{\mathrm{compat}}).$$

In this equation, r_i is observer patch i's local readout, $\mathcal{R}_i$ is its readout map, and $\{r_i\}_{\mathrm{compat}}$ is the compatible readout set. The map T repairs those records and returns the shared world. Left equation: subjective experience. Right equation: shared reality. Experience matters because OPH reality is the stable object made by compatible observer readouts [O1,O2]. Private incompatible records fail to become public records. The mechanism is mundane: records read, compare, repair, return.

Exact quantum event-surface hit

The fixed-cutoff record interface also emits textbook quantum readout. On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and the Lueders update on the OPH record surface. On the corresponding two-wing Bell event surface, the CHSH expression obeys

$$|S_{\mathrm{CHSH}}| \le 2\sqrt{2}.$$

This is an exact structural output of the finite record algebra. It uses no $\alpha(0)$, G, QCD, charged-lepton masses, or neutrino data [O1,O3,A18,A19].

2. Closure coordinates and counted witnesses

Standard physics carries roughly thirty independent numerical inputs, depending on how the count is made. OPH leaves a local pixel coordinate and a global screen-capacity coordinate. They are fixed by closure equations from the OPH rules [O1,O3,O4,O5,O6]. That is the zero-parameter claim. Observations may tell us which branch to inspect. They do not set the endpoint.

For the compact hierarchy row, write P_{src} for the selected pixel-branch coordinate consumed by the bridge certificate. The public endpoint coordinate is

$$P_C := \varphi + \alpha_{\mathrm{CODATA}}\sqrt{\pi} = 1.630968209403959\ldots.$$

For this compact row, P_{src} denotes that selected endpoint. The public endpoint is a branch locator only; it supplies no upstream source-map input. The source-side diagnostic candidate P_{cand} remains a parallel witness for the same formula, with the public endpoint excluded upstream. The hadron-completed endpoint is $P_{\rm full}$. It becomes a compact endpoint row only when the missing QCD/hadronic electromagnetic endpoint map is emitted by the OPH rules [O5,O6]. The global screen coordinate remains

$$N_\star \simeq N_{\rm scr}^{\rm dS} \simeq 3.31 \times 10^{122}.$$

In ordinary language, one elementary observer cell must read the same as simulated record content and simulating screen rule, and the universe must carry the record capacity required by the screen that contains that same universe.

Independent two-closure lock

The OPH universe is a self-reference. The simulating side supplies a readback rule, the simulated side supplies the record readout, and a consistent universe exists only when both sides close on the same object [O1,O2,O4]. The two closure equations are where this strange loop becomes a number. **The closure coordinates must be computed as OPH endpoints. The simulation is consistent only when the self-references hold.**

Local loop: try a pixel P_k. The OPH branch reads its field width $R_P(P_k)$. The pixel map turns that readout back into a corrected pixel:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}.$$

Solving $P = \Gamma_P(P)$ gives the simulating-side pixel setting for that branch. In the compact hierarchy row this coordinate is $P_{\rm src}$. The simulated-side field width comes from the same endpoint: $\alpha_{\rm src} = R_P(P_{\rm src})$.

Global loop: try a capacity N_k. The capacity map counts closed-screen normal forms, charges the screen budget, applies the Minimal Admissible Realization (MAR) selector, and returns the admissible capacity:

$$\Lambda_k = R_N(N_k), \qquad N_{k+1} = F_N(N_k).$$

Solving $N = F_N(N)$ gives the simulating-side capacity setting $N_\star$. The simulated-side horizon readout comes from the same computed endpoint: $\Lambda_\star = R_N(N_\star)$.

The solvers may start in intervals suggested by observation. The equations themselves must contain no measured α, measured Λ, weak masses, or calibrated target residuals. If $\Gamma_P : I_P \to I_P$ and $F_N : I_N \to I_N$ stay inside their declared intervals and shrink candidates, Banach gives one local endpoint and one global endpoint:

$$P_{\rm src} = \Gamma_P(P_{\rm src}), \qquad N_\star = F_N(N_\star).$$

Changing either endpoint breaks the resonance between the source setting and the universe read from inside [O1,O3,O4,O5].

The OPH screen branch supplies twelve curvature ports. Reversible write/check orientation doubles the product-adjoint repair register to 24. The local half of the lock appears as the 12-port exponent in the hierarchy proof below. This is where the fixed-point simulation stops being a metaphor. The pixel value and the screen capacity resonate: changing either one breaks the pixel-screen closure and removes the consistent universe. Measurements can label the basin; closure supplies the decimal.

The same endpoints have a limited carrier reading. $P_{\rm src}$ fixes a canonical geometric cell area, and $N_\star$ fixes total screen capacity. Under an equal-area screen chart the cell count is $K_{\rm cell} = 4N_\star/P_{\rm src}$. This does not select a fixed primitive observer algebra. The quantity $P_{\rm src}/4$ is shared cut entropy density, so it should not be read as $\log\dim\mathcal{H}_{\rm cell}$ for an independent cell factor. Variable finite federations remain the carrier reading unless a separate theorem selects an isomorphic one-cell or fixed-block primitive carrier.

The local constant is computed from a readback equation that does not use measured α [O3,O4,O5,O6]:

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\rm src}(P), \qquad \Gamma_{\rm src}(P) = \varphi + \frac{\sqrt{\pi}}{A_T^{\rm src}(P)}.$$

Read left to right: try a pixel value P, let the OPH branch emit its field widths, and let $\Gamma_{\rm src}$ return the corrected pixel value. The symbol P is a candidate identity value for one OPH pixel: the number that converts the cell's screen area into the edge entropy it shares with neighboring cells. The symbols $\alpha_U(P)$, $A_Z(P)$, and $A_T^{\rm src}(P)$ are the three OPH readouts along the way: unification width, electroweak anchor, and Ward-projected Thomson endpoint.

The public implementation solves the same loop in the α-chart [C1]. Start with a trial α_k. It defines a candidate pixel

$$P_k = \varphi + \alpha_k\sqrt{\pi}.$$

From P_k, the OPH branch computes the unification point, the electroweak anchor, and the Thomson endpoint. That returns the inner electromagnetic readout

$$\alpha_{\rm in}(P_k) = \frac{1}{A_T^{\rm src}(P_k)}.$$

The solver asks how far the returned α is from the trial α:

$$\rho(\alpha_k) = \alpha_{\rm in}(\varphi + \alpha_k\sqrt{\pi}) - \alpha_k.$$

The code scans the declared branch interval, finds where ρ crosses zero, and cuts the bracket in half until the mismatch is below the requested tolerance. In fixed-point notation, the same loop is

$$\alpha_{k+1} = \alpha_{\rm in}(\varphi + \alpha_k\sqrt{\pi}), \qquad P_{k+1} = \varphi + \alpha_{k+1}\sqrt{\pi}.$$

The compact hierarchy output is the pair $(P_{\rm src}, \alpha_{\rm src})$. CODATA never enters this map; it can only be printed after the solve as a comparison.

The simulating geometric reading and simulated electromagnetic reading are

$$\alpha_{\rm ext}(P) = \frac{P-\varphi}{\sqrt{\pi}}, \qquad \alpha_{\rm in}(P) = \frac{1}{A_T^{\rm src}(P)}, \qquad \alpha_{\rm ext}(P_{\rm src}) = \alpha_{\rm in}(P_{\rm src}),$$

The last equality is the pixel lock, equivalently $P_{\rm src} = \Gamma_{\rm src}(P_{\rm src})$. The golden ratio $\varphi = (1+\sqrt{5})/2$ is the geometric screen balance point. A purely golden pixel has geometry with no field-width readback. $\sqrt{\pi}$ converts boundary width into the pixel chart, and the electromagnetic width supplies the detuning. The low-energy electromagnetic width is read out as

$$\alpha_{\rm src}(0) = \frac{1}{A_T^{\rm src}(P_{\rm src})} = \frac{P_{\rm src} - \varphi}{\sqrt{\pi}}.$$

Solving the pixel equation determines $P_{\rm src}$ and the branch field width together. Approximate measured α may identify the branch interval; it never defines $A_T^{\rm src}$ or $\Gamma_{\rm src}$. At the fixed point one may write

$$P_{\rm src} = \varphi + \alpha_{\rm src}\sqrt{\pi}$$

as the readout identity of the solved loop.

Measured α is only a comparison

The equation is a self-reference with two outputs. The outer screen reading $\alpha_{\rm ext}(P) = (P - \varphi)/\sqrt{\pi}$ must equal the inner electromagnetic reading $\alpha_{\rm in}(P) = 1/A_T^{\rm src}(P)$ emitted by the same candidate pixel. Solving the loop gives both values. The fine-structure constant is an output of the same solve:

$$P_{\rm src} = \Gamma_{\rm src}(P_{\rm src}), \qquad \alpha_{\rm src} = \alpha_{\rm ext}(P_{\rm src}) = \alpha_{\rm in}(P_{\rm src}).$$

Reading the final identity backward with CODATA $\alpha(0)$ gives $P_C = \varphi + \alpha_{\rm CODATA}\sqrt{\pi}$. That is a useful comparison coordinate. The OPH calculation is the closure solve. The code path in [C1] keeps external comparison values out of the solver.

OPH has one major weakpoint: hadronic refinements

The weakpoint is precise. The compact standalone proof excludes the public Thomson endpoint from its core count. The pure OPH trunk emits $\alpha_{\rm cand}^{-1} = 136.9948351646\ldots$. The measured CODATA endpoint is $\alpha^{-1}(0) = 137.035999177(21)$. The OPH trunk value is too low by $\Delta_{\alpha^{-1}} = 0.0411640124\ldots$, about 3.0×10^{-4} relative. That miss is small on the scale of α^{-1}, but large compared with CODATA precision, so it cannot be hand-waved.

The parameter-free trunk value is reproducible from the public solver and checked runtime report [C1]:

```
cd code/P_derivation && python3 derive_p.py --no-hadron-closure
```

The saved report `runtime/full_p_alpha_report_current.json` prints $\alpha^{-1} = 136.99483516462164945794999458578 7\ldots$. The measured CODATA value never enters that command path [C1].

What is missing is the low-energy QCD/hadronic correction: confined quark and hadron effects in the electromagnetic endpoint. The missing term is a concrete finite-lattice QCD calculation that has to be emitted from the OPH rules. At the required resolution, the calculation is beyond conventional von Neumann hardware; OMEGA is the optical fixed-point compute lane for it [S14]. The source-side QCD map is absent from the compact proof. Public $\alpha(0)$ therefore stays in the empirical endpoint-closure lane [O5,O6].

The compact proof does not multiply the public $\alpha(0)$ endpoint, hadron masses, or the SI G clock row into its core count. It uses the QCD-free hierarchy witness below, which avoids the hadronic correction completely. $P_{\rm full}$ names the hadron-completed endpoint defined by that OPH QCD map.

The global constant is computed from the screen-capacity readback equation [O1,O4]:

$$N_\star = \mathcal{C}(N_\star).$$

The universe is allowed exactly the record capacity that its own closed-screen readback says it needs. The symbol N is a candidate total capacity for the closed observer screen: the number of record units the universe can carry in OPH units. $\mathcal{C}$ is the closed-screen readback rule. The simulated reading counts the closed-screen normal forms supported at capacity N. The simulating reading charges the screen for the N slots it claims. A self-contained fixed-point computation has no external memory bank, so the available records and the cost of the screen must close on the same capacity. The global loop says the simulated universe contains the capacity of the simulating screen that contains it [O1,O4]. The screen solver mirrors the pixel solver. Try a capacity N, count the closed-screen normal forms it supports, subtract the screen budget charge, and apply MAR to pick the admissible branch. The score is

$$S(N) = \log|\Omega_N^{\rm sc}| - N, \qquad N_\star = \mathrm{MAR}\arg\max_N S(N) \simeq 3.31 \times 10^{122}.$$

The set $\Omega_N^{\rm sc}$ contains the closed-screen normal forms at capacity N. The term $\log|\Omega_N^{\rm sc}|$ is the simulated-side count, and the subtraction of N is the simulating-side budget charge. The operator MAR is Axiom 5 of OPH [O1]: Minimal Admissible Realization. In the core papers, Axiom 5 says that the realized branch is the lexicographically minimal admissible representative under the branch complexity vector $C(\mathfrak{S})$. In this screen-capacity display, the same selector chooses the minimal admissible arithmetic representative of the maximizing branch. The output is the unique selected capacity. The cosmological constant comes from that same capacity:

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\rm cell} = \frac{3\pi P_{\rm src}}{N_\star}.$$

So $N_\star$ and $\Lambda_\star$ are computed together in the global closure, the same way $P_{\rm src}$ and $\alpha_{\rm src}$ are computed together in the local closure [O1,O4].

The three-scalar cross-lock

Before any decimal appears, the recovered compact-gauge branch fixes the Standard Model quotient

$$G_{\rm SM} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The derivation has a specific route. Overlap holonomy and zero-obstruction transport first build a compact bosonic gauge category. DR/Tannaka reconstruction reads off a compact group from that category. MAR then selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel of the realized action fix the hypercharge lattice and the $\mathbb{Z}_6$ quotient [O4,A21]. The same branch carries the protected massless photon, gluon, and graviton rows:

$$m_\gamma = 0, \qquad m_g = 0, \qquad m_{\text{grav}} = 0.$$

The same quotient fixes charge quantization: $Q = T_3 + Y$ and $Q_{\text{color singlet}} \in \mathbb{Z}$. Its connected adjoint is $(8,1,0)\oplus(1,3,0)\oplus(1,1,0)$. There are no mixed $(3,2,\pm 5/6)$ X/Y gauge bosons, so the ordinary simple-GUT gauge-mediated proton-decay channel is absent on this branch [O4,A20,A21]. This discrete structural witness sits upstream of the numerical rows. The same branch then emits the compact integer triple

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\text{EW}} = N_c + 1 = 4.$$

These three integers feed both compact numerical witnesses below. The same 4 controls the charged-family Koide phase and the weak hierarchy exponent [O3,O4,O5]. One source branch gives one huge hierarchy and a two-coordinate lepton-family shape.

The charged-family Koide front door. The charged-family row is the clean first hit: no QCD, no G, no public $\alpha(0)$, and no charged-lepton masses enter as inputs. On the realized Standard Model branch,

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\text{EW}} = N_c + 1 = 4.$$

The OPH-specific claim is the phase selector. A balanced three-family carrier would sit on the Koide cone for any phase; OPH fixes the phase from branch integers:

$$\delta_e = \frac{\beta_{\text{EW}}}{2N_c N_g} = \frac{2}{9}.$$

If R is the cyclic three-family shift, $R^3 = I$, define

$$C_{\delta_e} = I + \frac{1}{\sqrt{2}}\left(e^{i\delta_e}R + e^{-i\delta_e}R^2\right).$$

Its eigenvalues are

$$r_k = 1 + \sqrt{2}\cos\left(\delta_e + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

For $\delta_e = 2/9$, the ordered roots are

$$(r_e, r_\mu, r_\tau) = (0.040349908219207475,\ 0.5802119201475368,\ 2.3794381716332555).$$

The trace identities are immediate:

$$\operatorname{tr} C_{\delta_e} = 3, \qquad \operatorname{tr} C_{\delta_e}^2 = 6.$$

If charged square-root masses lie on this OPH carrier direction, $m_i = s_e r_i^2$, then Koide follows exactly:

$$\frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{s_e \operatorname{tr} C_{\delta_e}^2}{s_e (\operatorname{tr} C_{\delta_e})^2} = \frac{6}{9} = \frac{2}{3}.$$

The same direction predicts the scale-free charged-family ratios

$$\frac{m_\mu}{m_e} = \left(\frac{r_\mu}{r_e}\right)^2 = 206.770315972729\ldots,$$
$$\frac{m_\tau}{m_e} = \left(\frac{r_\tau}{r_e}\right)^2 = 3477.47283710460\ldots,$$
$$\frac{m_\tau}{m_\mu} = \left(\frac{r_\tau}{r_\mu}\right)^2 = 16.8180467333776\ldots.$$

Using the PDG lepton summary masses $m_e = 0.51099895000\,\text{MeV}$, $m_\mu = 105.6583755\,\text{MeV}$, and $m_\tau = 1776.86\,\text{MeV}$, the deviations are about 9.8 ppm, 70 ppm, and 60 ppm respectively [S8]. The Koide value computed from those same pole masses is about 9.2 ppm below 2/3. The mechanism is stronger than a Koide parametrization: OPH fixes the phase $\delta_e = 2/9$ from the same integer triple that appears again in the hierarchy exponent below. This row fails if N_c, N_g, or β_{EW} are not independently fixed by the Standard Model branch, if $\delta_e = 2/9$ is chosen from charged-lepton masses, or if the charged-family shape misses the sorted lepton direction [O5,O6,A15,A16].

The QCD-free hierarchy witness

OPH naturally solves the hierarchy problem. This is the second independent numerical witness from the same structural branch, and it also stays away from QCD. A route through the public Thomson endpoint, hadron masses, or the cesium clock would import the Ward-projected hadronic correction as an assumption. That criticism is fair, so this row avoids it. The OPH calculation fixes $\alpha_U(P_{\text{src}}) = 0.041124336195630495$ in the interval $I_U = [0.041123336195630494, 0.041125336195630496]$, with the Krawczyk image strictly inside I_U [O5]. The counted row is the weak-to-UV hierarchy emitted from P_{src} and that independently checked $\alpha_U(P_{\text{src}})$ value [O5,C3]. OPH gets the tiny weak scale from the two closure constants, without importing hadronic physics.

The factor 12 is screen topology doing the counting. On the triangulated S^2 screen, the locally flat triangular vacuum is six-valent. Define the curvature charge $q_v = 6 - \deg(v)$. Since $V - E + F = 2$ and $3F = 2E$, one has $E = 3V - 6$, hence

$$\sum_v q_v = 6V - \sum_v \deg(v) = 6V - 2E = 12.$$

A closed triangular screen cannot be flat everywhere. Exactly twelve units of screen curvature must appear. Refinement-stable MaxEnt spreads this total charge into twelve unit fivefold defects: at fixed $\sum q_v = 12$, the convex local defect cost $\sum q_v^2$ is minimized by unit charges, while negative defects require extra positive charge. Edge-center completion turns each defect collar into a central port [O3]. With no marked point, the twelve equal ports sit on the maximal finite rotation orbit A_5/C_5, whose size is $60/5 = 12$. Therefore the global screen depth $D = \log(N/\pi)$ is sampled locally as $D/12$.

The representation-to-spectrum certificate turns the same screen count into the oriented repair register:

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor two is reversible write/check orientation. Since $N = \pi R_{\mathrm{dS}}^2/\ell_\star^2$, the same $D = \log(N/\pi)$ is the cosmic length ratio in logarithmic form. The electroweak branch samples that global depth through the 12 unoriented ports and the $4P_{\mathrm{src}}$ projection below; the 24-slot register supplies the repair normalization [O4,O5,C3].

The local cell entropy is $\ell_{\mathrm{cell}} = P_{\mathrm{src}}/4$, and the electroweak projection coefficient is $\beta_{\mathrm{EW}} = N_c + 1 = 4$. Therefore

$$\Gamma_{\mathrm{EW}} = \beta_{\mathrm{EW}}\ell_{\mathrm{cell}}\frac{\log(N/\pi)}{12} = \frac{P_{\mathrm{src}}}{12}\log(N/\pi).$$

This is the local share of the global screen step: cell entropy, times the electroweak projection, times one twelfth of the screen depth. The compact standalone hierarchy witness is the transmutation law

$$\frac{v}{E_\star} = P_{\mathrm{src}}^{-1/2}\exp\left[-\frac{2\pi}{4\alpha_U(P_{\mathrm{src}})}\right] = 2.0199803239725553 \times 10^{-17}.$$

$E_\star$ is the OPH UV energy scale. $E_{\mathrm{cell}} = E_\star/\sqrt{P_{\mathrm{src}}}$ is the cell energy after the pixel area is accounted for. The exponent uses the OPH-emitted $\alpha_U(P_{\mathrm{src}})$, not G, Λ, M_W, M_Z, Higgs mass, the public Thomson endpoint, or a QCD hadronic endpoint [O5]. This is the QCD-free quantitative witness.

The bridge coordinate rewrites the same exponent as a screen-capacity readout:

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{bridge}}^{\mathrm{EW}}}{\pi}\right)^{-P_{\mathrm{src}}/12}, \qquad N_{\mathrm{bridge}}^{\mathrm{EW}} = \pi\exp\left[\frac{6\pi}{P_{\mathrm{src}}\alpha_U(P_{\mathrm{src}})}\right].$$

$N_{\mathrm{bridge}}^{\mathrm{EW}}$ is a derived capacity readout on the local/global hierarchy branch. On the public-endpoint pixel branch used by this compact certificate, the selected-branch proof closes it as a fixed point of the log-capacity map

$$\mathcal{C}_{\mathrm{EW}}(P,x) = (1-\lambda)x + \lambda\frac{6\pi}{P\alpha_U(P)}, \qquad \lambda = \frac{1}{2}.$$

For fixed $P = P_{\mathrm{src}}$, this is a contraction with unique endpoint

$$x_\star = \frac{6\pi}{P_{\mathrm{src}}\alpha_U(P_{\mathrm{src}})}, \qquad N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi e^{x_\star} = 3.5323546226929906\ldots \times 10^{122}.$$

The bridge residual

$$R_{\mathrm{EW}} := \alpha_U(P_{\mathrm{src}})^{-1} - \frac{P_{\mathrm{src}}}{6\pi}\log(N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi)$$

vanishes at that endpoint, equivalently

$$\alpha_U(P_{\mathrm{src}})\log(N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi) - \frac{6\pi}{P_{\mathrm{src}}} = 0.$$

The finite readback-resolution certificate supplies the positive-root readback limit

$$\rho_{\mathrm{read}}(r, N_{\mathrm{CRC}}^{\mathrm{EW}}) \longrightarrow (N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi)^{-1/2}.$$

Together with the 12-port screen sieve and the 24-slot oriented repair register, this closes the selected-branch local/global hierarchy resonance:

$$t_{\mathrm{tr}}(P_{\mathrm{src}}) = \frac{P_{\mathrm{src}}}{12}\log(N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi), \qquad \frac{v}{E_{\mathrm{cell}}} = (N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi)^{-P_{\mathrm{src}}/12}.$$

The rounded 3.31×10^{122} screen capacity remains the cosmological capacity display; the exact electroweak bridge uses the contraction endpoint above. No measured v, M_W, M_Z, m_H, G, Λ, or public $\alpha(0)$ enters as an upstream source-map input for this bridge certificate [O4,O5,C3].

Ordinary particle physics treats the Higgs scale as a tuning problem. In OPH the weak scale is a branch output with zero naturality defect:

$$\epsilon_H = 0, \qquad \epsilon_H \in [0,0].$$

Zero means there is no cancellation to arrange. There is no bare scalar dial to balance against a cutoff; the weak scale is the settled-form readout of the 24-slot oriented repair-register normalization. That is the hierarchy problem in standard language, with the standard fine-tuning term absent from the OPH normal form [O4,O5].

Discrete rigidity check

The integers $N_c = 3$, $N_g = 3$, and $\beta_{\mathrm{EW}} = 4$ are branch data with no extra measured decimals. The evidence is the overconstraint. Once the Standard Model branch fixes this integer triple, the same β_{EW} hits two unrelated readouts: the weak hierarchy and the charged-family shape. Hold the P_{src} and $\alpha_U(P_{\mathrm{src}})$ proof records fixed and change only β_{EW}:

β_{EW}	$v/E_\star$	m_μ/m_e	m_τ/m_e
3	5.97×10^{-23}	25.84	579.07
4	2.01998×10^{-17}	206.7703	3477.47
5	4.20×10^{-14}	outside cone	outside cone

The $\beta_{\mathrm{EW}} = 5$ charged carrier has

$$r_e = 1 + \sqrt{2}\cos\left(\frac{5}{18} + \frac{4\pi}{3}\right) = -0.0158500583\ldots,$$

so the light square-root mass leaves the positive cone. The nearby branch values miss both sectors at once. The 4 is therefore a positive-edge branch: it keeps the hierarchy in the observed sector and produces a tiny positive electron root without fitting charged masses [O4,O5,C3].

Prediction escrow: rows exposed to failure

Known constants can point to branch neighborhoods. They cannot carry the whole case. OPH has to freeze outputs that are not used by the Koide or hierarchy rows and then let external measurements attack them.

Neutrino weighted-cycle branch. The weighted-cycle branch emits

$$(m_1, m_2, m_3) = (0.017454720257976796,\ 0.019481987935919015,\ 0.05307522145074924)\ \text{eV},$$

with

$$\sum m_\nu = 0.09001192964464505\ \text{eV}, \quad \Delta m^2_{21} = 7.488059465106851 \times 10^{-5}\ \text{eV}^2,$$
$$\Delta m^2_{31} = 2.5123118727618473 \times 10^{-3}\ \text{eV}^2, \qquad (\alpha_{21}, \alpha_{31}) = (153.618518^\circ, 257.003241^\circ).$$

The same branch emits the PMNS readout

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (34.225904631810025^\circ,\ 49.72282845058266^\circ,\ 8.686355527700156^\circ),$$

$$\delta_{\text{PMNS}} = 305.58061231449796^\circ, \qquad J = -0.02753115613565372, \qquad \frac{\Delta m^2_{21}}{\Delta m^2_{32}} = 0.030721110097966534.$$

The same output gives direct laboratory and cosmology targets. With $c_{ij} = \cos\theta_{ij}$ and $s_{ij} = \sin\theta_{ij}$,

$$m_\beta = \sqrt{c^2_{12}c^2_{13}m^2_1 + s^2_{12}c^2_{13}m^2_2 + s^2_{13}m^2_3} = 0.0196244372655102\ \text{eV}.$$

For neutrinoless double beta decay, set $\alpha^{0\nu}_{31} = \alpha_{31} - 2\delta_{\text{PMNS}}$. The branch gives

$$m_{\beta\beta} = \left| c^2_{12}c^2_{13}m_1 + s^2_{12}c^2_{13}m_2 e^{i\alpha_{21}} + s^2_{13}m_3 e^{i\alpha^{0\nu}_{31}} \right| = 0.00797676286284509\ \text{eV}.$$

This row does not tune the hierarchy or charged-family shape. Oscillation, absolute-mass, cosmology, beta-endpoint, or neutrinoless-double-beta data can reject the branch. In the particle paper and code it is theorem-level on the declared weighted-cycle branch; the positive-segment adapter and bridge-coordinate readouts stay diagnostic and do not feed back into the theorem state [O5,C2,A17,S9,S10,S11].

Charged-family residual. The known Koide relation is background. The OPH claim is the branch phase and the residual. Define

$$\eta_e := \delta^{\text{phys}}_e - \frac{2}{9}.$$

The compact row uses $\delta_e = 2/9$ as a branch cross-lock with the hierarchy integer triple. A full charged-family promotion has to emit η_e and the affine determinant-line anchor from the same OPH branch [O5,O6,A15,A16].

Running-scheme rigidity. The hierarchy row counts only when the $\alpha_U(P_{\text{src}})$ proof record fixes the high-scale beta packet, threshold packet, representation cutoffs, matching prescription, and renormalization convention outside the target value. Tuning any of those with $v/E_\star$ as the target demotes the row to calibration. A stronger closure emits the running packet from the same OPH branch, or emits a threshold spectrum that experiments can exclude [O5,C3,A8,A10].

Row	Role in this note	Decisive check
$v/E_\star$	Measurement-free hierarchy witness	Frozen $\alpha_U(P_{\text{src}})$ packet with no measured-answer path.
$m_\mu/m_e,\ m_\tau/m_e$	Scale-free charged-family shape	OPH map for η_e and the affine determinant-line anchor.
ν masses, phases, and endpoint targets	Prediction escrow	External data accept or exclude the frozen branch.
$\alpha(0)$	Empirical endpoint closure	OPH hadronic spectral map emits the missing QCD correction.
G_{SI}	Clock-hierarchy display	No-G cesium gap is emitted without a gravity-calibrated proxy.

3. Geometric gravity normalization

OPH gets gravity before meters, kilograms, or seconds enter the discussion. G_{geom} is Newton coupling in OPH geometric units, a_{cell} is one observer-cell area, $\bar{\ell}_{\text{shared}}$ is its shared edge-entropy weight, and $\ell_\star$ is the OPH length quantum. The edge-entropy/area-law theorem on the OPH gravity branch identifies $a_{\text{cell}}/(4\bar{\ell}_{\text{shared}})$ with the structural coefficient of the stress-energy term in the local Einstein equation $G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}}\langle T_{ab}\rangle$ and in the Newton-Poisson weak-field limit $\nabla^2\Phi = 4\pi G_{\text{geom}}\rho$; the same G_{geom} propagates literally through the Jacobson chain into both limits [O4,A22]. On the realized local-pixel branch that theorem reduces to

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}, \qquad a_{\text{cell}} = P_{\text{src}}\ell^2_\star, \qquad \bar{\ell}_{\text{shared}} = \frac{P_{\text{src}}}{4}.$$

The A22 derivation fixes the factor 4. Substitution gives

$$\frac{G_{\text{geom}}}{\ell^2_\star} = \frac{P_{\text{src}}}{4(P_{\text{src}}/4)} = 1, \qquad G_{\text{geom}} = \ell^2_\star.$$

The same pixel number counts area and shared entropy. The pixel number cancels. Gravity is the area quantum in OPH units. The SI decimal needs the separate no-G clock proof [O4,A6]. The edge-entropy/area-law theorem itself uses no measured G, no Planck area, no measured Λ, and no gravity-calibrated clock scale [O4,A22]. In the same theorem, the linear-in-area scaling $\langle L_C\rangle \propto A(\partial C)$ is derived inside OPH from the Markov-collar/refinement-stability structure of the screen net, the OPH structural

Einstein-equation coefficient is then $\kappa_{\mathrm{OPH}} = 2\pi\, a_{\mathrm{cell}}/\bar{\ell}_{\mathrm{shared}}$, and the Newton-Poisson convention $\nabla^2\Phi = 4\pi G_N \rho$ defining the Newton coupling fixes $G_N = \kappa_{\mathrm{OPH}}/(8\pi) = a_{\mathrm{cell}}/(4\bar{\ell}_{\mathrm{shared}}) = G_{\mathrm{geom}}$. The factor of 4 is the ratio $\kappa_{\mathrm{OPH}}/(8\pi)$.

4. Compact proof rows

The compact argument counts only the rows that pass the anti-fitting rule: observer-patch architecture, the exact quantum event surface, the Standard Model structural branch, the QCD-free Koide shape witness, the QCD-free hierarchy witness, and the geometric gravity normalization. OPH claims to describe the whole machine, so the longer particle, cosmology, dark-sector, hardware, and SI rows matter. They stay outside this compact surface unless their own calculations pass the same rule [O1,O4,O5,C2,C3].

Rows outside the compact count remain useful evidence and falsification surfaces with their own claim classes. The readable falsifiability map is `extra/OPH_falsifiability.md`.

Witness	Counted here?	Why	Refs
Finite observer-patch normal form and patch-carrier pipeline	Yes	Finite observer records, overlaps, mismatch scores, repair maps, checkpoints, and patch carriers define the fixed-point architecture. The row tests the mechanism before any decimal fit enters.	[O1,O2,C5]
Born-Lueders and Tsirelson event surface	Yes	The finite central record algebra gives $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, Lueders conditioning, and $\lvert S_{\mathrm{CHSH}}\rvert \le 2\sqrt{2}$ on the declared Bell event surface.	[O1,O3,A18,A19]
Standard Model structural branch	Yes	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$, $N_c = 3$, $N_g = 3$, protected massless carriers, charge quantization, and no X/Y gauge bosons are discrete structural outputs. The same integers feed the hierarchy and Koide witnesses.	[O3,O4,O5,A20,A21]
$\alpha_U(P_{\mathrm{src}})$ interval check	Yes	The unified-coupling row is checked by an interval/Krawczyk calculation that does not read measured electroweak masses, low-energy couplings, G, Planck units, or Λ.	[O5,C2,A8]
Charged-family Koide shape	Yes, shape only	$\delta_e = (N_c+1)/(2N_cN_g) = 2/9$ and $r_k = 1 + \sqrt{2}\cos(\delta_e + 2\pi k/3)$ give $Q_{\mathrm{Koide}} = 2/3$. This fixes the scale-free charged-family direction at ppm contact with PDG charged-lepton ratios. Absolute lepton masses stay outside this counted row.	[O5,O6,A15,A16]
QCD-free hierarchy law	Yes	$v/E_\star = P_{\mathrm{src}}^{-1/2}\exp[-2\pi/(4\alpha_U(P_{\mathrm{src}}))]$. The weak scale is emitted without the QCD/hadronic endpoint and $\epsilon_H = 0$.	[O5,C3,A10]
Geometric gravity normalization	Yes, normalization	$G_{\mathrm{geom}} = \ell_\star^2$. This fixes the OPH area normalization. It is not the SI Newton-constant decimal.	[O3,O4,A7]
Public $\alpha(0)$ and P_C	No	The public endpoint uses the missing QCD/hadronic electromagnetic transport. $P_C = \varphi + \alpha_{\mathrm{CODATA}}\sqrt{\pi}$ is a comparison coordinate under the declared missing-map condition.	[O5,O6,A13]
SI G	No	The SI decimal counts only when the no-G cesium clock hierarchy emits $\varepsilon_{\mathrm{Cs}}$ without measured G or gravity-calibrated proxies.	[O4,A6]
Particle, dark-sector, neutrino, cosmology, and collider rows	No	These rows remain useful tests with their declared claim classes. They are not multiplied into the compact anti-circularity argument.	[O5,O7,O8,C2,C4]
PIXEL, IBM, OMEGA, and hardware lanes	No	Hardware lanes are experimental surfaces unless their public bundles supply task definitions, raw readouts, controls, manifests, and verifier logs.	[S12,S13,S14]

The compact rows in this table are derived from OPH closure coordinates and the OPH axioms, with the full derivation chains and proofs contained in the referenced papers. Comparison and test rows keep their declared evidence class and failure conditions. For details, see the linked papers and the public code in `github.com/FloatingPragma/observer-patch-holography`.

5. Falsifiers and evidence classes

This note does not assign a numerical probability to OPH. That would be unrealistic. OPH outputs the basic structure of our universe, compresses the necessary constants to two unique fixed points, and turns the hard physics problems below into consequences of one closure grammar. A random fit has to work much harder than this.

Physics problems that become easy to solve with OPH

One fitted constant is cheap. This claim has a harder job. It has to hit the compact rows and make the observer problem, gravity/gauge reconstruction, hierarchy, dark energy, dark matter, Yang-Mills gap, unification without proton decay, particle inventory, and string vacuum selection fall out of the same two fixed points. String theory earned attention because its quantized strings naturally contain a spin-2 graviton. OPH produces the protected massless spin-2 branch in the observer-overlap reconstruction, and the string-vacuum paper shows why string theory is an effective description of the same controlled compact branch [O4,O9,S15].

Observer problem: standard physics has no clean public-world role for subjective experience. OPH makes experience a finite readout boundary, and public reality is the compatible fixed point of those readouts [O1,O2].

Gravity with gauge physics: standard physics keeps general relativity and gauge theory in separate formalisms. OPH puts Lorentz kinematics, the $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ three-dimensional spatial chart, the Jacobson-Einstein branch, compact gauge reconstruction, the Standard Model quotient, hypercharge, three colors, and three generations in one observer-overlap reconstruction stack [O3,O4,O5].

Hierarchy problem: OPH naturally solves the hard Higgs naturality problem. The 12-port screen sieve, the 24-slot oriented repair register, and the selected local/global capacity bridge emit the weak scale with $\epsilon_H = 0$, so the usual fine-tuning term disappears [O4,O5,C3].

Yang-Mills gap: OPH turns the Clay-style gap problem into a repair gap statement. On a controlled compact-gauge branch, after the continuum certificate in [O10] is supplied, the Yang-Mills mass gap is $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}} > 0$. Without that certificate, finite repair remains regulator-level gap accounting [O4,O10].

Dark energy: OPH turns the cosmological-constant problem into global screen bookkeeping. The cosmological constant is the closed-screen capacity readout of the same Einstein branch. Arbitrary vacuum-energy input has no role in that row [O1,O4].

Dark matter: OPH replaces an unseen-particle assumption by repair-stress bookkeeping on its declared dark branch. The public dark-sector rows have continuation and scorecard labels; full likelihood and calculation chain closure is their separate test. The static galaxy continuation gives the functional target $\nu_{\mathrm{OPH}}(x) = [1 - \exp(-\lambda_{\mathrm{collar}}\sqrt{x})]^{-1}$, with $g_{\mathrm{obs}} \simeq \sqrt{a_{\mathrm{eff}} g_b}$ and $v^4 = GM_b a_{\mathrm{eff}}$ in the deep low-acceleration limit [O7,O8,C4].

Unification without proton decay: OPH gets unification-style running without the classic simple-GUT proton-decay problem. The edge branch runs unification-style while the realized product quotient has no simple-GUT leptoquark generators [O4,O5].

Particle inventory: the compact proof counts structural carriers, gauge quotient data, the charged-family Koide shape witness, the $\alpha_U(P)$ interval check, and the QCD-free hierarchy witness. W/Z, absolute charged-lepton masses, selected-frame quarks, Higgs/top, and neutrino rows remain in the particle files with their declared support levels [O4,O5,C2].

String vacuum selection: the same observer-patch constraints define a vacuum sieve over critical-string candidates. The string-vacuum note reads strings as an effective edge language and selects the Bouchard-Donagi witness only behind the critical-edge, cohomology, safety-layer, threshold, and moduli-locking gates. This is the precise sense in which OPH recovers string theory as an effective description inside the OPH stack; no ordinary string vacuum is promoted to the OPH-native vacuum without those certificates [O9,S15].

Tests outside the compact proof count

The next entries are clean ways to break OPH branches. They sit outside the compact proof until their own calculation and checks are supplied without reading the measured answer.

Conditional screen-spectrum and CMB lift [O2,O4]. The compact/main paper states the finite-screen theorem as a conditional continuation: scalar readout and inverse-repair receipts can promote the declared branch to a MaxEnt Gaussian screen covariance and the

$$\theta_{\mathrm{scr}} = P/48$$

screen exponent, but not by themselves to a primordial curvature amplitude or TT/TE/EE spectrum. Only after the complete bridge passes may the thin-shell power-law branch write $n_s = 1 - \theta_{\mathrm{scr}}$. The source-side bridge is

$$q_\zeta = \frac{1}{3}\log\frac{J_X}{\bar{J}_X} = \zeta_\rho, \qquad \mathcal{L}_u\zeta_a = -\frac{\Theta}{3(\rho+p_{\mathrm{eff}})}\Gamma_a^{\mathrm{eff}},$$

with

$$\Gamma_a^{\mathrm{eff}} = 0 + \text{ repair gap} + \text{ rank-one mode} \Longrightarrow \text{primordial curvature}$$

only after the radial prior, null-space, phase-coherence, isocurvature, and forward-residual receipts pass. The missing steps are explicit gates: scalar fixed expected quadratic release energy, radial screen-to-primordial lift, a finite covariant collar-packet parent for any dark/anomaly source, Boltzmann transfer, source-provenance, pooled global reducers, and frozen likelihood execution. The promoted quantities $\eta_R, A_\zeta, q_{\mathrm{IR}}, \ell_{\mathrm{IR}}, B_A, \rho_A$, any certified Γ_{rec}, and N_{CRC} must come from an immutable source dependency DAG; contradictory no-data-use metadata fails closed, shard-local nonlinear averages are diagnostics, and N_{CRC} is the consensus invariant unless a disjoint additive capacity schema is supplied. A scalar $\rho_A(a), B_A(k,a)$ table and a transition spectral number are not enough: the latter remains $\gamma_{\text{repair step}}$ until the active fiber, physical clock, and common-parent response pole are certified. If the exchange branch is nonzero, the source packet must also carry explicit recipient stress and equal-and-opposite exchange current, and all source/solver/likelihood hashes must be immutable before data comparison. The consensus-protocol paper supplies the relevant certificate language and fixed-point firewall; this compact proof cites that package instead of reproducing the full math [A23,A24,A25].

Collider tests [O5,C2,S6]. The OPH particle files contain concrete collider-facing numbers, including $m_H = 125.1995304097179\,\mathrm{GeV}$, $m_t = 172.35235532883115\,\mathrm{GeV}$, $M_W = 80.377\,\mathrm{GeV}$, and $M_Z = 91.18797809193725\,\mathrm{GeV}$, with the declared scheme and evidence caveats in [O5,C2]. OPH also predicts no low-scale visible gauge branch incompatible with the recovered $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$ quotient [O4,O5]. Falsification: scheme-controlled collider measurements exclude the stated mass surfaces, or CERN/FCC discovers a light visible gauge boson or chiral exotic that cannot live on the OPH quotient branch.

χ_ν lift response [O8,O7,C4]. The supporting row listed in [O8] uses the public comparison coordinate P_C: $\chi_\nu^{\mathrm{can}} = e^{-P_C/24} = 0.9343006394893864\ldots$, with $0.9343006394893864 \le \chi_\nu^{\mathrm{can}} \le 1$, and $\chi_\nu^{\mathrm{eng}} = \chi_\nu^{\mathrm{can}}/N_{\mathrm{coh}}$. For $N_{\mathrm{coh}} = 10^{22}$, this is $9.34 \times 10^{-23} \le \chi_\nu^{\mathrm{eng}} \le 10^{-22}$. A hoverboard-class areal load $\Sigma = 200$ to $600\,\mathrm{kg\,m^{-2}}$ needs a vertical canonical coherence contrast $\Delta S_{\mathrm{coh}}^{\mathrm{can}} \simeq 2 \times 10^{-8}$ to 6×10^{-8} for full response. Test: build a substrate that logs ΔS_{coh}, $A_\perp$, Σ, force, temperature, acoustic drive, EM pickup, and mechanical coupling. A null run gives

$$|\chi_\nu^{\mathrm{eng}}| \le \frac{4\pi G F_{\min}}{g^2 A_\perp |\Delta S_{\mathrm{coh}}^{\mathrm{eng}}|}.$$

Falsification: "hoverboard doesn't work" after the declared contrast is produced and the control channels stay null.

Dark-sector cosmology [O7,C4]. The diagnostic branch records $\Omega_A = 0.264114401$, $\Omega_m^{\mathrm{CAMB}} = 0.315905206$, $\sigma_8 = 0.807787204$, $S_8 = 0.828924037$, and $\rho_A/\rho_b = 5.363470441$. Test: first emit a finite covariant collar-packet parent with stress closure, recipient stress and exchange-current closure for any nonzero exchange branch, gauge-independent B_A, active-fiber/physical-clock receipts for any promoted Γ_{rec}, refinement convergence, and frozen likelihood hashes; then run the public dark-sector scorecard in [C4] and replace the compact CAMB/SPARC scaffolds by the full covariance likelihoods named in [O7]. Falsification: the fixed branch fails the joint CMB, BAO, SPARC, cluster, and weak-lensing likelihood.

Neutrino closure [O5,C2]. The prediction-escrow section freezes the weighted-cycle masses, splittings, and phases. Test: combine cosmological mass-sum constraints, beta endpoint data, oscillation fits, and neutrinoless-double-beta limits. Falsification: the absolute mass scale or phase-sensitive data exclude the branch. Fastest kill tests are listed in the falsifiability map: $\sum m_\nu = 0.0900119296\,\mathrm{eV}$, beta endpoint $m_\beta \simeq 0.0196\,\mathrm{eV}$, $\Delta m^2_{21} = 7.4880594651 \times 10^{-5}\mathrm{eV}^2$, $\Delta m^2_{32} = 2.4374312781 \times 10^{-3}\mathrm{eV}^2$, the listed mixing angles, and $\delta = 305.581°$.

In this note, we show that the output universe of an Observer Patch Holography simulation looks suspiciously like our universe. The fourteen-paper OPH stack cannot be printed inside a short note; that is what the papers, code, and empirical evidence notes in the OPH repository are for: `github.com/FloatingPragma/observer-patch-holography`. What we can say here is that OPH emits the slice shown above, and the same stack extends the pattern to observer records, quantum event surfaces, spacetime, gravity, gauge structure, charge quantization, the hierarchy, dark energy, dark matter, particle structure, and falsifiable neutrino targets. Given that evidence, it seems very unlikely that we are not observer patches in an OPH simulation. A critic can kill the compact proof by finding a path from a measured answer into a counted calculation, breaking a uniqueness or interval check, or falsifying the Koide shape or QCD-free hierarchy witness. Supporting rows have their own tests and their own claim classes. Every counted row says exactly what is being counted.

Appendix A. Assigning a number to Newton's constant

The geometric gravity row in the compact argument is

$$G_{\mathrm{geom}} = \ell_\star^2.$$

The SI decimal is a separate clock translation. It counts only when the cesium gap is emitted without measured G, Planck units, or gravity-calibrated proxies [O4,A6]:

$$\varepsilon_{\mathrm{Cs}}^{\mathrm{calc}} = \mathcal{H}_{\mathrm{Cs}}\left(\alpha, \frac{m_e}{E_\star}, \frac{\Lambda_{\mathrm{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\mathrm{atom}}^{\mathrm{Cs}}\right).$$

If $\varepsilon_{\mathrm{Cs}}$ is back-solved from measured G, the row is a display check. With an independently emitted clock gap, exact SI constants c, $\hbar$, and $\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{s}^{-1}$ give [S3,S4]

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2\hbar\nu_{\mathrm{Cs}}^2}\varepsilon_{\mathrm{Cs}}^2 = 6.674299995910528\ldots \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}.$$

The finite-lattice QCD/hadronic part of the cesium calculation is infeasible on conventional von Neumann hardware at the required resolution. OMEGA targets that fixed-point compute class [S14]. This appendix is included for orientation and is not counted in the compact anti-circularity proof.

References

Data and external references

[S1] CODATA value for Newton's constant G.
[S2] CODATA inverse fine-structure constant α^{-1}.
[S3] CODATA exact speed of light c.
[S4] CODATA reduced Planck constant $\hbar$.
[S5] Planck Collaboration cosmological parameters.
[S6] Particle Data Group gauge and Higgs-boson summary tables.
[S7] Particle Data Group quark summary tables.
[S8] Particle Data Group lepton summary tables.
[S9] NuFIT 6.0 neutrino oscillation fit.
[S10] KATRIN direct neutrino-mass result.
[S11] Particle Data Group neutrino-mass review.
[S12] PIXEL optical verification plan.
[S13] IBM quantum-cloud OPH experiment code and raw results.
[S14] OMEGA optical fixed-point compute hardware.
[S15] The Right String Theory, OPH blog reference writeup.
[S16] Douglas R. Hofstadter, *I Am a Strange Loop*, Basic Books.

OPH papers

[O1] Observers Are All You Need.
[O2] Reality as Consensus Protocol.
[O3] Screen Microphysics and Observer Synchronization.
[O4] Recovering Relativity and Standard Model Structure from Observer Overlap Consistency.
[O5] Deriving the Particle Zoo from Observer Consistency.
[O6] Fine-Structure Constant Derivation.
[O7] OPH Dark Matter Paper.
[O8] χ_ν Susceptibility Bounds.
[O9] Observer Patch Holography as String Vacuum Selector.
[O10] Yang-Mills Gap and the Clay Problem.

Code and supporting data

[C1] Pixel-constant derivation code.
[C2] Particle-branch code and supporting files.
[C3] Hierarchy certificates and local/global resonance code.
[C4] Dark-sector scorecard and continuation code.
[C5] Consensus and fixed-point architecture code.

Direct reading anchors

Fast lookup: axioms [A1], consensus repair [A2], simulation criterion [A23], consensus refinement/firewall [A24], conditional screen-spectrum gate [A25], pixel closure [A4], capacity closure [A5], hierarchy witness [A10], Standard Model quotient [A21].

[A1] OPH axioms [O1:L328-334].
[A2] Consensus and repair normal form [O2:L534-550].
[A3] Patch-carrier and screen microphysics [O3:L150-166].
[A4] Pixel fixed-point theorem form [O6:L895-922].
[A5] Capacity fixed point $N_\star$ [O4:L4146-4200].
[A6] No-G clock requirement [O4:L276-313].
[A7] Scale certificate and Newton-coupling display [O4:L580-661].
[A8] Unified-coupling interval check $\alpha_U(P)$ [O5:L1538-1559].
[A9] Electroweak bridge boundary [O4:L416-442].
[A10] Hierarchy/naturality proof stack [O5:L1566-1991].
[A11] 12-port screen sieve [O5:L1848-1895].

[**A12**] Fine-structure trunk [O5:L1458-1477].

[**A13**] Public endpoint caveat [O6:L1018-1024].

[**A14**] Support boundary [O4:L5388-5400].

[**A15**] Koide charged-family roots [O6:L451-459].

[**A16**] Charged-family shape boundary [O5:L2601-2672].

[**A17**] Neutrino weighted-cycle theorem branch [O5:L642-644].

[**A18**] Born-Lueders central-record theorem [O3:L384-398].

[**A19**] Bell/CHSH Tsirelson event surface [O1 fragment:L1078-1095].

[**A20**] Product-group consequences and no X/Y proton-decay carrier [O4:L4919-4941].

[**A21**] Standard Model quotient and $\mathbb{Z}_6$ kernel [O4:L4561-4917].

[**A22**] Edge-entropy/area law identifies the Einstein/Newton coupling and derives the factor of 4 [O4:L3869-4045].

[**A23**] OPH simulation certificate, fixed-point firewall, and falsifiers [O2: `def:oph-simulation`, `thm:oph-simulation`, `rem:oph-simulation-falsifiers`].

[**A24**] Finite and refinement consensus theorem package, including quotient closure, coarse-graining compatibility, Markov-collar splice, and the certificate boundary used by downstream conditional continuations [O2].

[**A25**] Conditional OPH screen-spectrum theorem and firewall: screen covariance/tilt are conditional; scalar release amplitude, source provenance, pooled reducers, radial lift, and physical TT/TE/EE spectra remain separate gates [O4: `thm:compact-conditional-screen-spectrum`].